

Polynomial — Grade 6 Ready Type Examples

One Easy, Medium, and Hard sample per Ready Grade 6 type, formatted like printable worksheets (shared instructions, graphs, and diagrams).

Rectangles with fraction side lengths

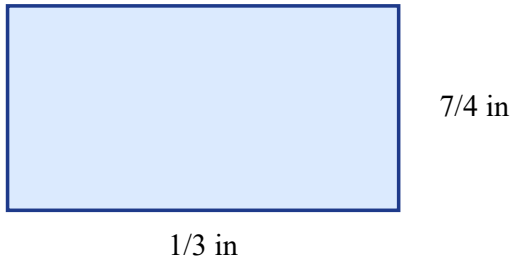
g6_rectangles_with_fraction_side_lengths · Grade 6 – Area and Volume with Fractions ·
gen=g6_fraction_rectangle_area

Name: _____ Date: _____

Find the area.

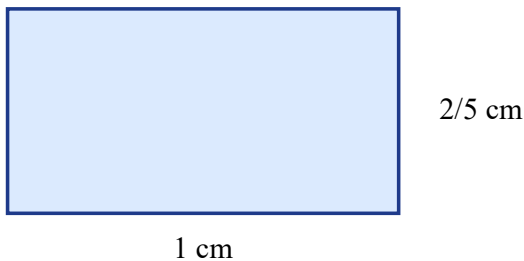
EASY

1. Find the area of the rectangle with base $\frac{1}{3}in$ and height $\frac{7}{4}in$.



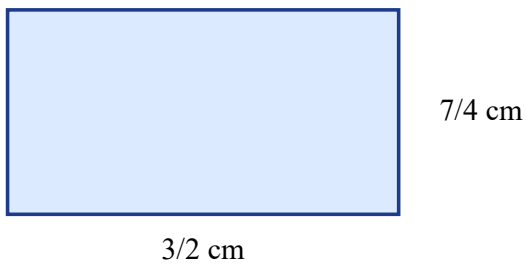
MEDIUM

2. Find the area of the rectangle with base $1cm$ and height $\frac{2}{5}cm$.



HARD

3. Find the area of the rectangle with base $\frac{3}{2}cm$ and height $\frac{7}{4}cm$.



Answer Key

1. **Easy:** $\frac{7}{12}in^2$
2. **Medium:** $\frac{2}{5}cm^2$
3. **Hard:** $\frac{21}{8}cm^2$

Right rectangular prisms with fraction side lengths

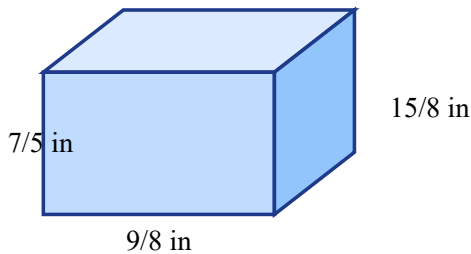
g6_right_rectangular_prisms_with_fraction_side_lengths · Grade 6 – Area and Volume with Fractions ·
gen=g6_fraction_prism_volume

Name: _____ Date: _____

Find the volume.

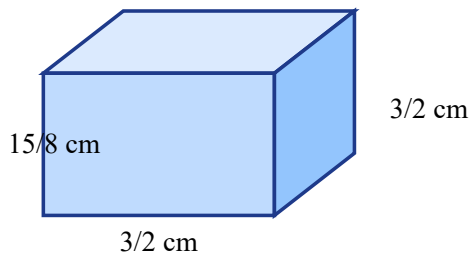
EASY

1. Find the volume of the right rectangular prism with side lengths $\frac{9}{8}in$, $\frac{15}{8}in$, $\frac{7}{5}in$.



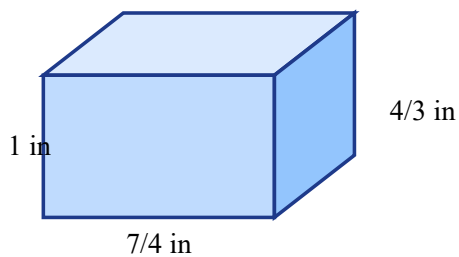
MEDIUM

2. Find the volume of the right rectangular prism with side lengths $\frac{3}{2}cm$, $\frac{3}{2}cm$, $\frac{15}{8}cm$.



HARD

3. Find the volume of the right rectangular prism with side lengths $\frac{7}{4}in$, $\frac{4}{3}in$, $1in$.



Answer Key

1. **Easy:** $\frac{189}{64}in^3$
2. **Medium:** $\frac{135}{32}cm^3$
3. **Hard:** $\frac{7}{3}in^3$

Triangles with fraction side lengths

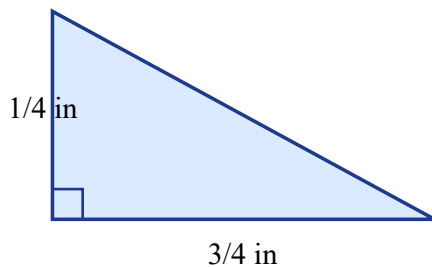
g6_triangles_with_fraction_side_lengths · Grade 6 – Area and Volume with Fractions ·
gen=g6_fraction_triangle_area

Name: _____ Date: _____

Find the area.

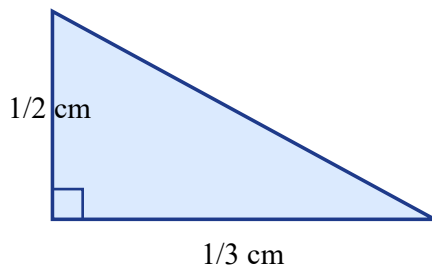
EASY

1. Find the area of the right triangle with base $\frac{3}{4}$ in and height $\frac{1}{4}$ in.



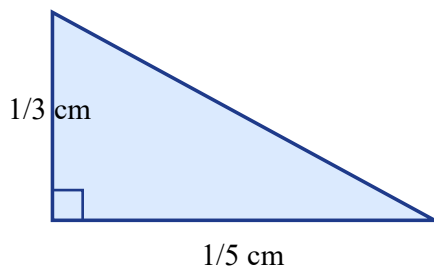
MEDIUM

2. Find the area of the right triangle with base $\frac{1}{3}$ cm and height $\frac{1}{2}$ cm.



HARD

3. Find the area of the right triangle with base $\frac{1}{5}cm$ and height $\frac{1}{3}cm$.



Answer Key

1. **Easy:** $\frac{3}{32}in^2$
2. **Medium:** $\frac{1}{12}cm^2$
3. **Hard:** $\frac{1}{30}cm^2$

Factoring

g6_factoring · Grade 6 – Common Factors and Common Multiples · gen=g6_factoring

Name: _____ Date: _____

Factor.

EASY

1. Write the prime factorization of 35

MEDIUM

2. Write the prime factorization of 10

HARD

3. Write the prime factorization of 153

Answer Key

1. **Easy:** $5 \cdot 7$

2. **Medium:** $2 \cdot 5$

3. **Hard:** $3^2 \cdot 17$

GCF and LCM word problems

g6_gcf_and_lcm_word_problems · Grade 6 – Common Factors and Common Multiples ·
gen=g6_gcf_and_lcm_word_problems

Name: _____ Date: _____

Solve the problem.

EASY

1. You have 12 cookies and 16 cookies.
What is the greatest number of identical bags you can make?

MEDIUM

2. Event A happens every 4 minutes and Event B every 17 minutes. After how many minutes do they happen together?

HARD

3. You have 44 cookies and 55 cookies. What is the greatest number of identical bags you can make?

Answer Key

1. **Easy:** 4
2. **Medium:** 68
3. **Hard:** 11

Greatest common factor

g6_greatest_common_factor · Grade 6 – Common Factors and Common Multiples ·
gen=g6_greatest_common_factor

Name: _____ Date: _____

Find the GCF.

EASY

1. Find the GCF of 24, 20

MEDIUM

2. Find the GCF of 55, 11

HARD

3. Find the GCF of 72, 108, 9

Answer Key

1. **Easy:** 4

2. **Medium:** 11

3. **Hard:** 9

Least common multiple

g6_least_common_multiple · Grade 6 – Common Factors and Common Multiples · gen=g6_least_common_multiple

Name: _____ Date: _____

Find the LCM.

EASY

1. Find the LCM of 28, 4, 22

MEDIUM

2. Find the LCM of 53, 56, 48

HARD

3. Find the LCM of 57, 60

Answer Key

1. **Easy:** 308

2. **Medium:** 17808

3. **Hard:** 1140

Distances on the coordinate plane

g6_distances_on_the_coordinate_plane · Grade 6 – Coordinate Plane · gen=geo_coordinate_distance

Name: _____ Date: _____

Find the distance.

EASY

1. Find the distance between $(-2, -4)$ and $(-2, 3)$.

MEDIUM

2. Find the distance between $(4, -8)$ and $(4, 6)$.

HARD

3. Find the distance between $(-6, 2)$ and $(-6, -9)$.

Answer Key

1. **Easy:** 7

2. **Medium:** 14

3. **Hard:** 11

Distances, word problems

g6_coordinate_plane_distances_word_problems · Grade 6 – Coordinate Plane · gen=wp_coordinate_distance

Name: _____ Date: _____

Solve the problem.

EASY

1. Point A is at $(-2, -3)$ and point B is at $(4, 1)$ on a coordinate plane. How far apart are the points in units?

MEDIUM

2. Point A is at $(3, 5)$ and point B is at $(8, 12)$ on a coordinate plane. How far apart are the points in units?

HARD

3. Point A is at $(-3, 2)$ and point B is at $(4, 10)$ on a coordinate plane. How far apart are the points in units?

Answer Key

1. **Easy:** 7
2. **Medium:** 9
3. **Hard:** 11

Points on the coordinate plane

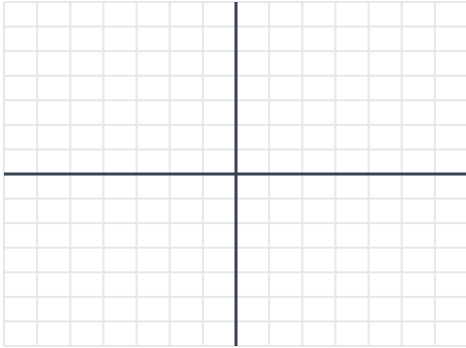
g6_points_on_the_coordinate_plane · Grade 6 – Coordinate Plane · gen=plotting_points

Name: _____ Date: _____

Identify the point.

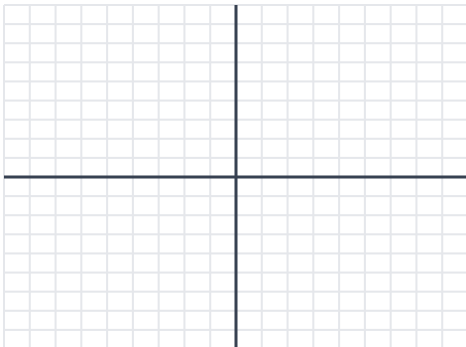
EASY

1. $(0, -5)$



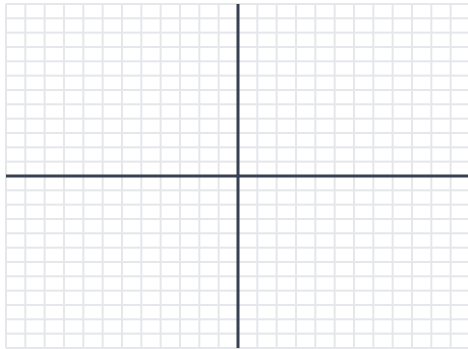
MEDIUM

2. $(-1, -7)$



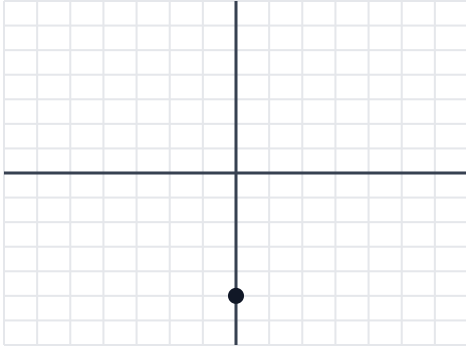
HARD

3. $(5, -4)$

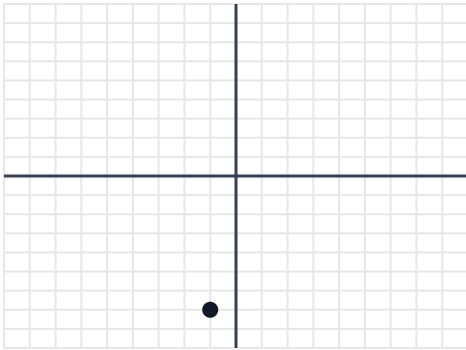


Answer Key

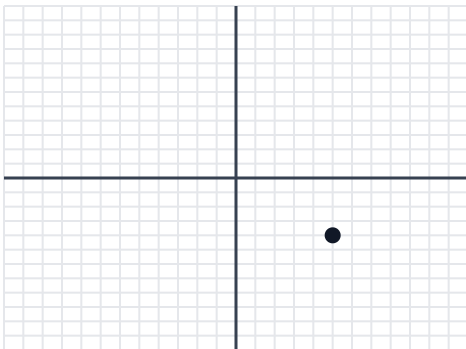
1. **Easy:** $(0, -5)$



2. **Medium:** $(-1, -7)$



3. **Hard:** $(5, -4)$



Shapes and perimeter on the coordinate plane

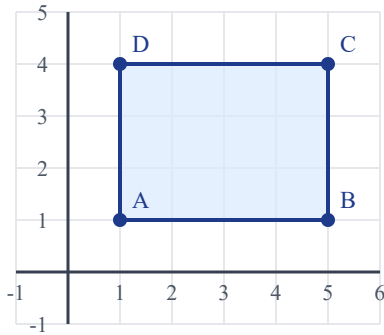
g6_shapes_and_perimeter_on_the_coordinate_plane · Grade 6 – Coordinate Plane ·
gen=g6_coordinate_perimeter

Name: _____ Date: _____

Find the perimeter.

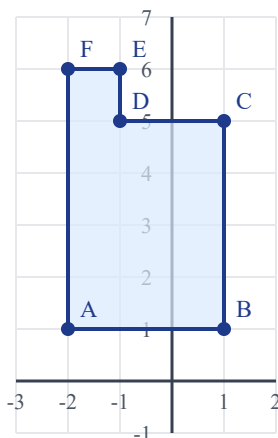
EASY

1. A rectangle on the coordinate plane has vertices $A(1, 1)$, $B(5, 1)$, $C(5, 4)$, $D(1, 4)$. Find the perimeter.



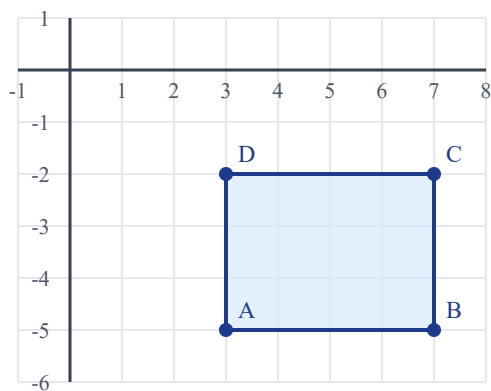
MEDIUM

2. A L-shaped polygon on the coordinate plane has vertices $A(-2, 1)$, $B(1, 1)$, $C(1, 5)$, $D(-1, 5)$, $E(-1, 6)$, $F(-2, 6)$. Find the perimeter.



HARD

3. A rectangle on the coordinate plane has vertices $A(3, -5)$, $B(7, -5)$, $C(7, -2)$, $D(3, -2)$. Find the perimeter.



Answer Key

1. **Easy:** 14
2. **Medium:** 16
3. **Hard:** 14

Center and spread

g6_data_center_and_spread · Grade 6 – Data Sets and Distributions · gen=stats_center_spread

Name: _____ Date: _____

Find the measure.

EASY

1. Find the median of the data set: $\{16, 4, 1, 7, 14, 14, 15, 5\}$.

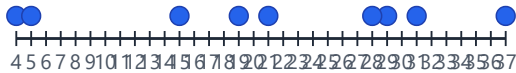
Data set



MEDIUM

2. Find the range of the data set: $\{31, 29, 15, 21, 28, 19, 4, 37, 5\}$.

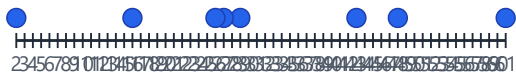
Data set



HARD

3. Find the median of the data set: $\{29, 2, 27, 26, 61, 48, 43, 16\}$.

Data set



Answer Key

1. **Easy:** 10.5
2. **Medium:** 33
3. **Hard:** 28

Drawing dot plots

g6_drawing_dot_plots · Grade 6 – Data Sets and Distributions · gen=g6_drawing_dot_plot

Name: _____ Date: _____

Create a dot plot for the data.

EASY

1. Create a dot plot for the data set $\{2, 7, 12, 9, 3, 2, 8, 9\}$. Use the blank axis provided.

Dot plot



MEDIUM

2. Create a dot plot for the data set $\{1, 6, 4, 6, 4, 12\}$. Use the blank axis provided.

Dot plot



HARD

3. Create a dot plot for the data set $\{9, 2, 9, 10, 10, 9\}$. Use the blank axis provided.

Dot plot



Answer Key

1. **Easy:** (see figure)
2. **Medium:** (see figure)
3. **Hard:** (see figure)

Drawing histograms

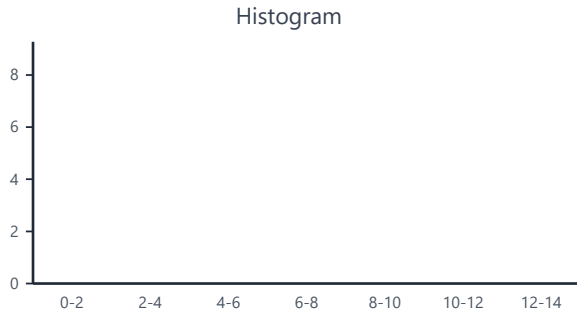
g6_drawing_histograms · Grade 6 – Data Sets and Distributions · gen=g6_drawing_histogram

Name: _____ Date: _____

Create a histogram for the data.

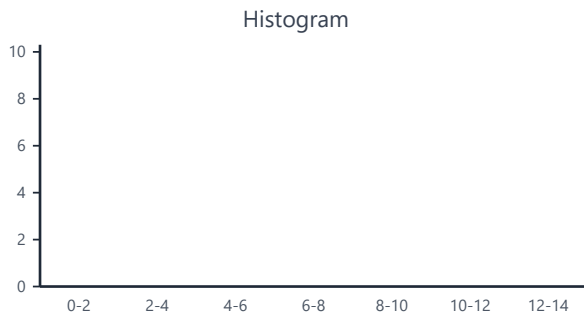
EASY

1. Create a histogram for the data set $\{4, 11, 6, 6, 1, 6, 2, 2, 12\}$. Use the blank axis provided.



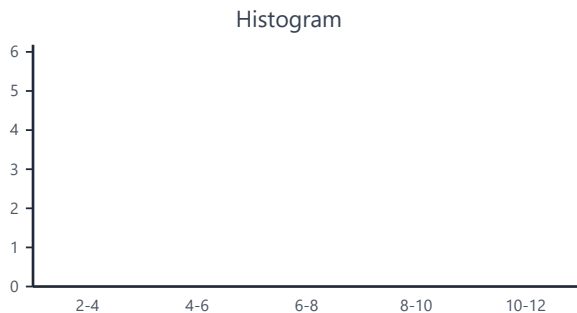
MEDIUM

2. Create a histogram for the data set $\{4, 1, 3, 5, 9, 9, 11, 10, 11, 9\}$. Use the blank axis provided.



HARD

3. Create a histogram for the data set $\{10, 10, 6, 2, 2, 9\}$. Use the blank axis provided.



Answer Key

1. **Easy:** (see figure)
2. **Medium:** (see figure)
3. **Hard:** (see figure)

Interpreting box plots

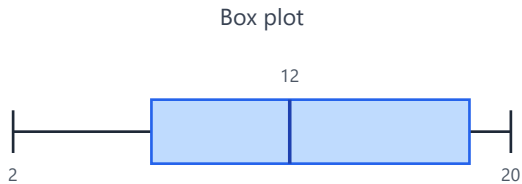
g6_interpreting_box_plots · Grade 6 – Data Sets and Distributions · gen=stats_box_plot_basics

Name: _____ Date: _____

Interpret the box plot.

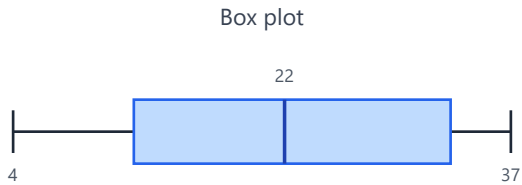
EASY

1. From the box plot, find the range.



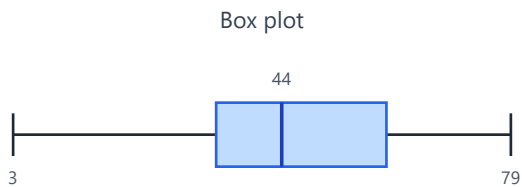
MEDIUM

2. From the box plot, find the range.



HARD

3. From the box plot, find the median.



Answer Key

1. **Easy:** 18

2. **Medium:** 33

3. **Hard:** 44

Interpreting dot plots

g6_interpreting_dot_plots · Grade 6 – Data Sets and Distributions · gen=stats_dot_plot_read

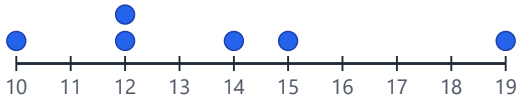
Name: _____ Date: _____

Interpret the dot plot.

EASY

1. The dot plot shows the data set below. How many dots are at 15?
Data: {19, 15, 10, 12, 14, 12}

Dot plot



MEDIUM

2. The dot plot shows the data set below. How many dots are at 5?
Data: {40, 32, 15, 7, 4, 5, 11, 4, 3, 4, 17, 11}

Dot plot



HARD

3. The dot plot shows the data set below. How many dots are at 12?
Data: {34, 25, 12, 68, 24, 62, 77, 77, 61}

Dot plot



Answer Key

1. **Easy:** 1

2. **Medium:** 1

3. **Hard:** 1

Interpreting histograms

g6_interpreting_histograms · Grade 6 – Data Sets and Distributions · gen=stats_histogram_read

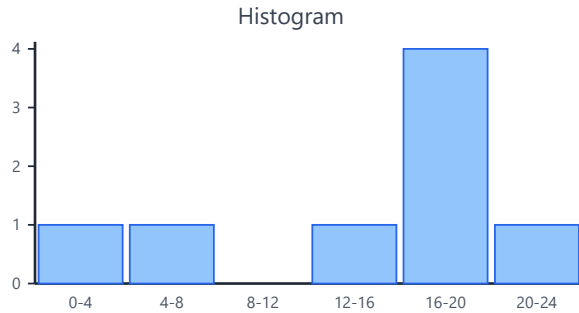
Name: _____ Date: _____

Interpret the histogram.

EASY

1. How many values fall in the interval $[4, 8)$?

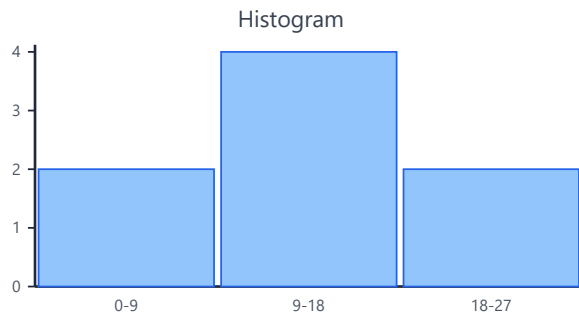
Data: $\{19, 13, 16, 19, 1, 20, 4\}$



MEDIUM

2. How many values fall in the interval $[18, 27)$?

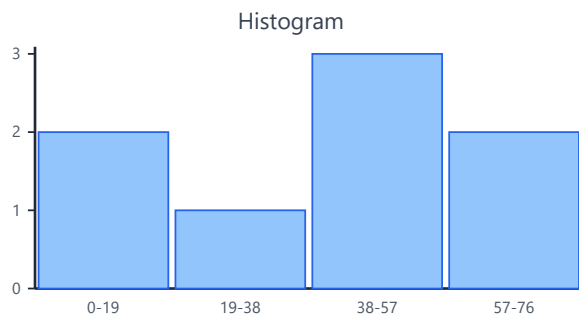
Data: $\{12, 4, 16, 8, 22, 15, 17, 23\}$



HARD

3. How many values fall in the interval $[0, 19)$?

Data: $\{39, 3, 21, 13, 60, 54, 64, 43\}$



Answer Key

1. Easy: 1

2. Medium: 2

3. Hard: 2

Decimal addition

g6_decimal_addition · Grade 6 – Decimal Arithmetic · gen=g6_decimal_addition

Name: _____ Date: _____

Add.

EASY

1. $88.3 + 62.6$

MEDIUM

2. $50.97 + 96.52$

HARD

3. $-58.233 - 37.466$

Answer Key

1. **Easy:** 150.9

2. **Medium:** 147.49

3. **Hard:** -95.699

Decimal multiplication

g6_decimal_multiplication · Grade 6 – Decimal Arithmetic · gen=g6_decimal_multiplication

Name: _____ Date: _____

Multiply.

EASY

1. $19 \cdot 0.9$

MEDIUM

2. $2.86 \cdot 4.14$

HARD

3. $4.6 \cdot 2.16$

Answer Key

1. **Easy:** 17.1

2. **Medium:** 11.8404

3. **Hard:** 9.936

Decimal multiplication with equivalent fractions

g6_decimal_multiplication_with_equivalent_fractions · Grade 6 – Decimal Arithmetic ·
gen=g6_decimal_multiplication

Name: _____ Date: _____

Multiply.

EASY

1. $20 \cdot 0.6$

MEDIUM

2. $6.68 \cdot 2.6$

HARD

3. $2.9 \cdot 7.9$

Answer Key

1. **Easy:** 12

2. **Medium:** 17.368

3. **Hard:** 22.91

Decimal subtraction

g6_decimal_subtraction · Grade 6 – Decimal Arithmetic · gen=g6_decimal_subtraction

Name: _____ Date: _____

Subtract.

EASY

1. $98.9 - 18.5$

MEDIUM

2. $71.35 - 35.02$

HARD

3. $42.108 - 38.552$

Answer Key

1. **Easy:** 80.4

2. **Medium:** 36.33

3. **Hard:** 3.556

Dividing decimals by decimals

g6_dividing_decimals_by_decimals · Grade 6 – Decimal Arithmetic · gen=g6_decimal_divide

Name: _____ Date: _____

Divide.

EASY

1. $1.5 \div 0.5$

MEDIUM

2. $12.0 \div 1.5$

HARD

3. $2.0 \div 0.5$

Answer Key

1. **Easy:** 3

2. **Medium:** 8

3. **Hard:** 4

Dividing decimals by whole numbers

g6_dividing_decimals_by_whole_numbers · Grade 6 – Decimal Arithmetic ·
gen=g6_dividing_decimals_by_whole_numbers

Name: _____ Date: _____

Divide.

EASY

1. $51.3 \div 9$

MEDIUM

2. $55.2 \div 8$

HARD

3. $19.585 \div 5$

Answer Key

1. **Easy:** 5.7

2. **Medium:** 6.9

3. **Hard:** 3.917

Dividing whole numbers by decimals

g6_dividing_whole_numbers_by_decimals · Grade 6 – Decimal Arithmetic · gen=g6_whole_by_decimal_divide

Name: _____ Date: _____

Divide.

EASY

1. $4 \div 0.2$

MEDIUM

2. $6 \div 0.5$

HARD

3. $14 \div 0.2$

Answer Key

1. **Easy:** 20

2. **Medium:** 12

3. **Hard:** 70

Dividing whole numbers that result in decimals

g6_dividing_whole_numbers_that_result_in_decimals · Grade 6 – Decimal Arithmetic ·
gen=g6_dividing_whole_numbers_that_result_in_decimals

Name: _____ Date: _____

Divide.

EASY

1. $2 \div 10$

MEDIUM

2. $35 \div 4$

HARD

3. $104 \div 16$

Answer Key

1. **Easy:** 0.2

2. **Medium:** 8.75

3. **Hard:** 6.5

Long division with remainders

g6_long_division_with_remainders · Grade 6 – Decimal Arithmetic · gen=g6_long_division_with_remainders

Name: _____ Date: _____

Divide. Write the quotient and remainder.

EASY

1. $83 \div 5$

MEDIUM

2. $766 \div 4$

HARD

3. $9123 \div 34$

Answer Key

1. **Easy:** 16 R 3

2. **Medium:** 191 R 2

3. **Hard:** 268 R 11

Dividing fractions

g6_dividing_fractions · Grade 6 – Dividing Fractions · gen=g6_fraction_divide

Name: _____ Date: _____

Divide.

EASY

1. $\frac{1}{2} \div \frac{1}{5}$

MEDIUM

2. $\left(-\frac{10}{9}\right) \div \left(\frac{5}{6}\right)$

HARD

3. $\left(-\frac{4}{7}\right) \div \left(\frac{1}{16}\right)$

Answer Key

1. **Easy:** $\frac{5}{2}$

2. **Medium:** $-\frac{4}{3}$

3. **Hard:** $-\frac{64}{7}$

How many groups/times?

g6_how_many_groups_times · Grade 6 – Dividing Fractions · gen=g6_fraction_divide_groups

Name: _____ Date: _____

Solve the problem.

EASY

1. How many groups of 6 are in 30?

MEDIUM

2. How many groups of $1\frac{1}{4}$ are in 5?

HARD

3. How many groups of $1\frac{3}{4}$ are in $10\frac{1}{2}$?

Answer Key

1. **Easy:** 5

2. **Medium:** 4

3. **Hard:** 6

How much in each group/time?

g6_how_much_in_each_group_time · Grade 6 – Dividing Fractions · gen=g6_fraction_divide_each

Name: _____ Date: _____

Solve the problem.

EASY

1. If 48 is shared equally into 6 groups, how much is in each group?

MEDIUM

2. If 6 is shared equally into 4 groups, how much is in each group?

HARD

3. If $7\frac{3}{11}$ is used in $1\frac{9}{11}$ of a period, how much is used in one full period?

Answer Key

1. **Easy:** 8
2. **Medium:** $1\frac{1}{2}$
3. **Hard:** 4

What fraction of a whole?

g6_what_fraction_of_a_whole · Grade 6 – Dividing Fractions · gen=g6_fraction_of_whole

Name: _____ Date: _____

Solve the problem.

EASY

1. What fraction of a whole is $\frac{1}{2}$ of $\frac{1}{3}$?

MEDIUM

2. What fraction of a whole is $\frac{1}{2}$ of $\frac{5}{11}$?

HARD

3. What fraction of a whole is $\frac{1}{2}$ of $\frac{19}{2}$?

Answer Key

1. **Easy:** $\frac{1}{6}$

2. **Medium:** $\frac{5}{22}$

3. **Hard:** $\frac{19}{4}$

Equations word problems

g6_equations_word_problems · Grade 6 – Equations · gen=wp_one_step_equation

Name: _____ Date: _____

Solve the problem.

EASY

1. Hazel collected 4 equal donations totaling \$52. How much was each donation?

MEDIUM

2. Yasmin collected 5 equal donations totaling \$85. How much was each donation?

HARD

3. Evan collected 2 equal donations totaling \$18. How much was each donation?

Answer Key

1. Easy: \$13
2. Medium: \$17
3. Hard: \$9

Hanger diagrams

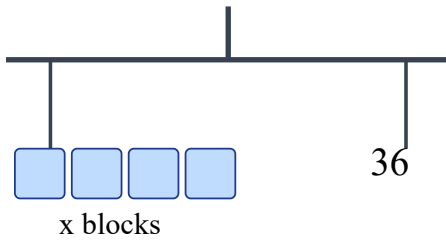
g6_equations_hanger_diagrams · Grade 6 – Equations · gen=g6_equations_hanger_diagrams

Name: _____ Date: _____

Solve the problem.

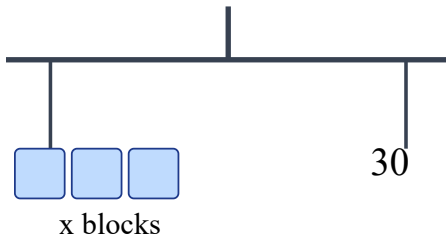
EASY

1. Use the diagram to solve $4x = 36$.



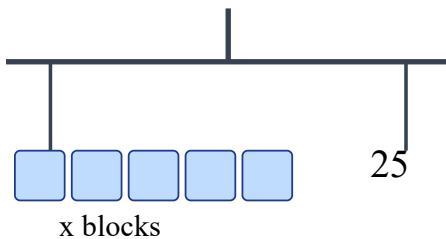
MEDIUM

2. Use the diagram to solve $3x = 30$.



HARD

3. Use the diagram to solve $5x = 25$.



Answer Key

1. **Easy:** 9

2. **Medium:** 10

3. **Hard:** 5

Solutions to equations

g6_solutions_to_equations · Grade 6 – Equations · gen=check_equation_solution

Name: _____ Date: _____

Determine if the value is a solution.

EASY

1. Is $x = 2$ a solution of $2x + 15 = 17$?

MEDIUM

2. Is $x = 8$ a solution of $6x + 11 = 77$?

HARD

3. Is $x = 12$ a solution of $5x + 9 = 19$?

Answer Key

1. **Easy:** no

2. **Medium:** no

3. **Hard:** no

Tape diagrams

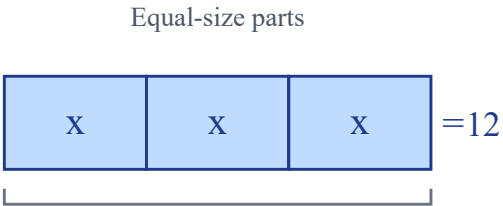
g6_equations_tape_diagrams · Grade 6 – Equations · gen=g6_equations_tape_diagrams

Name: _____ Date: _____

Solve the problem.

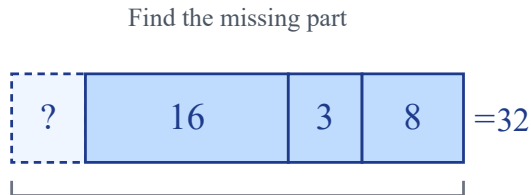
EASY

1. Use the tape diagram to find x .



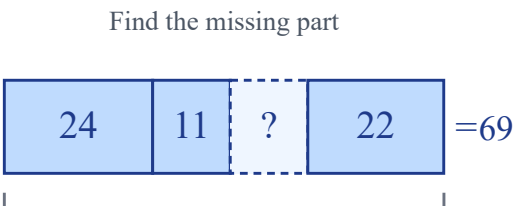
MEDIUM

2. Use the tape diagram to find the missing value.



HARD

3. Use the tape diagram to find the missing value.



Answer Key

1. **Easy:** 4

2. **Medium:** 5

3. **Hard:** 12

Constant rate equations

g6_constant_rate_equations · Grade 6 – Equations as Relationships between Two Variables ·
gen=write_one_step_equation

Name: _____ Date: _____

Write the equation.

EASY

1. A bike travels at 10 miles per hour.
Write an equation for the distance d
after 3 hours, then find d .

MEDIUM

2. A bike travels at 5 miles per hour.
Write an equation for the distance d
after 7 hours, then find d .

HARD

3. A bike travels at 3 miles per hour. Write an equation for the distance d after 4
hours, then find d .

Answer Key

1. **Easy:** $d = 10 \cdot 3$; $d = 30$
2. **Medium:** $d = 5 \cdot 7$; $d = 35$
3. **Hard:** $d = 3 \cdot 4$; $d = 12$

Equations for other relationships

g6_equations_for_other_relationships · Grade 6 – Equations as Relationships between Two Variables ·
gen=write_one_step_equation

Name: _____ Date: _____

Write the equation.

EASY

1. A bike travels at 10 miles per hour.
Write an equation for the distance d
after 8 hours, then find d .

MEDIUM

2. A bike travels at 10 miles per hour.
Write an equation for the distance d
after 4 hours, then find d .

HARD

3. A bike travels at 3 miles per hour. Write an equation for the distance d after 7
hours, then find d .

Answer Key

1. **Easy:** $d = 10 \cdot 8$; $d = 80$
2. **Medium:** $d = 10 \cdot 4$; $d = 40$
3. **Hard:** $d = 3 \cdot 7$; $d = 21$

Equivalent ratio equations

g6_equivalent_ratio_equations · Grade 6 – Equations as Relationships between Two Variables ·
gen=solving_proportions

Name: _____ Date: _____

Write the equation.

EASY

1. $\frac{x}{6} = \frac{1}{6}$

MEDIUM

2. $\frac{2}{9} = \frac{x}{45}$

HARD

3. $\frac{x}{3} = \frac{4}{3}$

Answer Key

1. **Easy:** 1

2. **Medium:** 10

3. **Hard:** 4

Combining like terms

g6_combining_like_terms · Grade 6 – Equivalent Expressions · gen=g6_combining_like_terms

Name: _____ Date: _____

Simplify.

EASY

1. $1x + 3x + 0$

MEDIUM

2. $3x + 7x + 3$

HARD

3. $9x + 22x + 12x + 1x + 9x + 11x + 2x + 11x$

Answer Key

1. **Easy:** $4x$

2. **Medium:** $10x + 3$

3. **Hard:** $77x$

Distributive property with area diagrams, algebraic

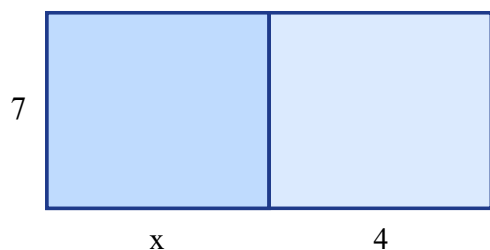
g6_distributive_property_area_diagrams_algebraic · Grade 6 – Equivalent Expressions ·
gen=g6_area_model_algebraic

Name: _____ Date: _____

Simplify.

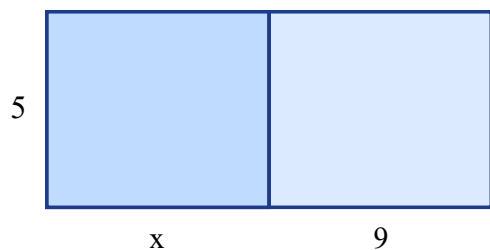
EASY

1. Use the area model to expand $7(x + 4)$.



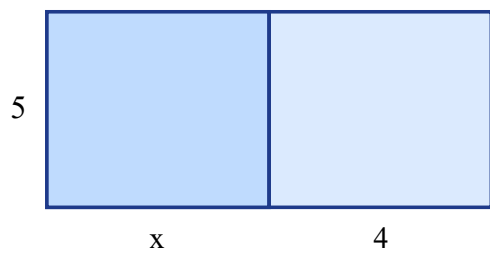
MEDIUM

2. Use the area model to expand $5(x + 9)$.



HARD

3. Use the area model to expand $5(x + 4)$.



Answer Key

1. **Easy:** $7x + 28$

2. **Medium:** $5x + 45$

3. **Hard:** $5x + 20$

Distributive property, algebraic

g6_distributive_property_algebraic · Grade 6 – Equivalent Expressions ·
gen=distributive_property_algebraic

Name: _____ Date: _____

Simplify.

EASY

1. $4(x + 3)$

MEDIUM

2. $6(x + 4)$

HARD

3. $7(x + 1)$

Answer Key

1. **Easy:** $4x + 12$

2. **Medium:** $6x + 24$

3. **Hard:** $7x + 7$

Hanger diagrams

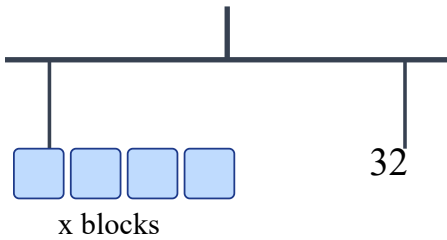
g6_inequalities_hanger_diagrams · Grade 6 – Inequalities · gen=g6_inequalities_hanger_diagrams

Name: _____ Date: _____

Solve the problem.

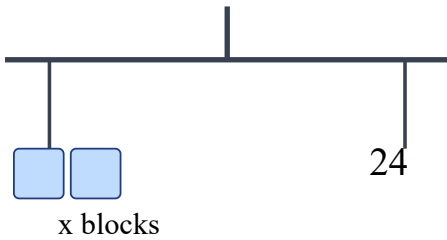
EASY

1. Use the diagram to solve $4x \leq 32$.



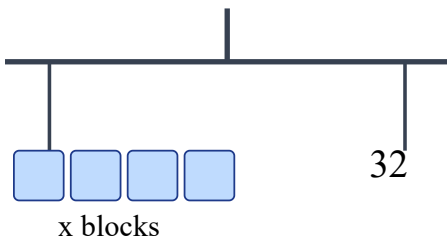
MEDIUM

2. Use the diagram to solve $2x \leq 24$.



HARD

3. Use the diagram to solve $4x \leq 32$.



Answer Key

1. **Easy:** $x \leq 8$

2. **Medium:** $x \leq 12$

3. **Hard:** $x \leq 8$

Inequalities word problems

g6_inequalities_word_problems · Grade 6 – Inequalities · gen=wp_inequality

Name: _____ Date: _____

Solve the problem.

EASY

1. Maria has scored a total of 130 points on 5 tests and wants at least 145 points after one more test. What is the minimum score needed on the next test?

MEDIUM

2. Paul has scored a total of 80 points on 4 tests and wants at least 113 points after one more test. What is the minimum score needed on the next test?

HARD

3. Lena has scored a total of 138 points on 5 tests and wants at least 157 points after one more test. What is the minimum score needed on the next test?

Answer Key

1. **Easy:** 15
2. **Medium:** 33
3. **Hard:** 19

Solutions to inequalities

g6_solutions_to_inequalities · Grade 6 – Inequalities · gen=graph_single_variable_inequality

Name: _____ Date: _____

Graph the following on the number line.

EASY

1. $x \leq 4$



MEDIUM

2. $x < -7$



HARD

3. $x < 16$



Answer Key

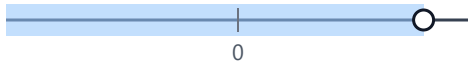
1. Easy: $x \leq 4$



2. Medium: $x < -7$



3. Hard: $x < 16$



Solving and graphing one-step inequalities

g6_solving_and_graphing_one_step_inequalities · Grade 6 – Inequalities · gen=one_step_inequalities

Name: _____ Date: _____

Solve and graph on the number line.

EASY

1. $x + 2 \geq 2$



MEDIUM

2. $3x \geq 30$



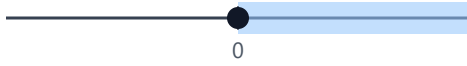
HARD

3. $\frac{x}{13} \geq -\frac{5}{26}$



Answer Key

1. **Easy:** $x \geq 0$



2. **Medium:** $x \geq 10$



3. **Hard:** $x \geq -\frac{5}{2}$



Writing and graphing inequalities

g6_writing_and_graphing_inequalities · Grade 6 – Inequalities · gen=graph_single_variable_inequality

Name: _____ Date: _____

Write and graph on the number line.

EASY

1. $x \geq 0$



MEDIUM

2. $x > 5$



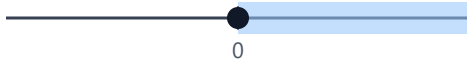
HARD

3. $x > 8$



Answer Key

1. **Easy:** $x \geq 0$



2. **Medium:** $x > 5$



3. **Hard:** $x > 8$



Absolute values

g6_absolute_values · Grade 6 – Negative Numbers and Absolute Value · gen=g6_absolute_values

Name: _____ Date: _____

Find the absolute value.

EASY

1. $|6|$

MEDIUM

2. $|-1|$

HARD

3. $|4|$

Answer Key

1. **Easy:** 6

2. **Medium:** 1

3. **Hard:** 4

Comparing numbers

g6_comparing_numbers · Grade 6 – Negative Numbers and Absolute Value · gen=g6_comparing_numbers

Name: _____ Date: _____

Compare the numbers.

EASY

1. -1 ? $-\frac{4}{9}$

MEDIUM

2. 12 ? $-\frac{2}{3}$

HARD

3. 1.87 ? -25

Answer Key

1. **Easy:** $<$

2. **Medium:** $>$

3. **Hard:** $>$

Comparing with absolute values

g6_comparing_with_absolute_values · Grade 6 – Negative Numbers and Absolute Value ·
gen=g6_comparing_with_absolute_values

Name: _____ Date: _____

Compare the values.

EASY

1. $|-5|$? $|-6|$

MEDIUM

2. $|19|$? $|17|$

HARD

3. $|31|$? $|34|$

Answer Key

1. **Easy:** $<$

2. **Medium:** $>$

3. **Hard:** $<$

Number line word problems

g6_number_line_word_problems · Grade 6 – Negative Numbers and Absolute Value · gen=wp_number_line

Name: _____ Date: _____

Solve the problem.

EASY

1. The temperature was -15°F and dropped 13°F . What is the new temperature?

MEDIUM

2. The temperature was -14°F and dropped 3°F . What is the new temperature?

HARD

3. The temperature was -8°F and rose 19°F . What is the new temperature?

Answer Key

1. **Easy:** -28
2. **Medium:** -17
3. **Hard:** 11

Numbers on a number line

g6_numbers_on_a_number_line · Grade 6 – Negative Numbers and Absolute Value · gen=number_line_plot

Name: _____ Date: _____

Plot the following numbers on the number line.

EASY

1. 6



MEDIUM

2. 10



HARD

3. -1



Answer Key

1. Easy: 6



2. Medium: 10



3. Hard: -1



Opposites of numbers

g6_opposites_of_numbers · Grade 6 – Negative Numbers and Absolute Value · gen=g6_opposites_of_numbers

Name: _____ Date: _____

Find the opposite.

EASY

1. Find the opposite of 8

MEDIUM

2. Find the opposite of 4

HARD

3. Find the opposite of -35

Answer Key

1. **Easy:** -8

2. **Medium:** -4

3. **Hard:** 35

Ordering numbers

g6_ordering_numbers · Grade 6 – Negative Numbers and Absolute Value · gen=g6_ordering_numbers

Name: _____ Date: _____

Order the numbers.

EASY

1. Order from least to greatest: 10 , $\frac{7}{10}$, 7

MEDIUM

2. Order from least to greatest: -3.5 , -3.37 , -13

HARD

3. Order from least to greatest: -3.74 , $-\frac{39}{8}$, $-\frac{9}{7}$

Answer Key

1. **Easy:** $\frac{7}{10}$, 7 , 10

2. **Medium:** -13 , -3.5 , -3.37

3. **Hard:** $-\frac{39}{8}$, -3.74 , $-\frac{9}{7}$

Ordering with absolute values

g6_ordering_with_absolute_values · Grade 6 – Negative Numbers and Absolute Value ·
gen=g6_ordering_with_absolute_values

Name: _____ Date: _____

Order the values.

EASY

1. Order from least to greatest absolute value: $-2, 9, -5$

MEDIUM

2. Order from least to greatest absolute value: $8, 4, 0$

HARD

3. Order from least to greatest absolute value: $41, 3, 32$

Answer Key

1. Easy: $-2, -5, 9$
2. Medium: $0, 4, 8$
3. Hard: $3, 32, 41$

Distributive property, numeric

g6_distributive_property_numeric · Grade 6 – Numeric Expressions, Exponents, and the Order of Operations · gen=distributive_property

Name: _____ Date: _____

Evaluate.

EASY

1. $2(8 - 5)$

MEDIUM

2. $8(9 - 1)$

HARD

3. $3(1 - 1)$

Answer Key

1. **Easy:** 6

2. **Medium:** 64

3. **Hard:** 0

Numeric expressions and the order of operations

g6_numeric_expressions_and_order_of_operations · Grade 6 – Numeric Expressions, Exponents, and the Order of Operations · gen=order_of_operations

Name: _____ Date: _____

Evaluate.

EASY

1. $4 \div 2 \cdot 1$

MEDIUM

2. $(8 + 6) \cdot 6 - 2$

HARD

3. $6 + 6 \cdot 3^2$

Answer Key

1. **Easy:** 2

2. **Medium:** 82

3. **Hard:** 60

Numeric expressions with exponents

g6_numeric_expressions_with_exponents · Grade 6 – Numeric Expressions, Exponents, and the Order of Operations · gen=g6_numeric_expressions_with_exponents

Name: _____ Date: _____

Evaluate.

EASY

1. $4 + 3 \cdot 2^2$

MEDIUM

2. $2^3 \div 4$

HARD

3. $3^3 + 2 \cdot 6$

Answer Key

1. **Easy:** 16

2. **Medium:** 2

3. **Hard:** 39

Properties of addition and multiplication

g6_properties_of_addition_and_multiplication · Grade 6 – Numeric Expressions, Exponents, and the Order of Operations · gen=g6_properties_of_addition_and_multiplication

Name: _____ Date: _____

Identify the property.

EASY

1. $5 \cdot 0 = 0$

- A. identity property of multiplication
- B. associative property of addition
- C. commutative property of multiplication
- D. zero property of multiplication

MEDIUM

2. $0 + 6 = 6$

- A. commutative property of multiplication
- B. commutative property of addition
- C. identity property of addition
- D. distributive property

HARD

3. $(4 \cdot 10) \cdot 4 = 4 \cdot (10 \cdot 4)$

- A. associative property of multiplication
- B. commutative property of multiplication
- C. associative property of addition
- D. identity property of addition

Answer Key

- 1. **Easy:** D · zero property of multiplication
- 2. **Medium:** C · identity property of addition
- 3. **Hard:** A · associative property of multiplication

Writing numeric expressions

g6_writing_numeric_expressions · Grade 6 – Numeric Expressions, Exponents, and the Order of Operations
· gen=writing_numeric_expressions

Name: _____ Date: _____

Write the expression.

EASY

1. Write an expression for 3 more than 5.

MEDIUM

2. Write an expression for the quotient of the sum of 6 and 12, and 5.

HARD

3. Write an expression for 10 more than the product of 5 and 5.

Answer Key

1. **Easy:** $5 + 3$

2. **Medium:** $(6 + 12) \div 5$

3. **Hard:** $5 \cdot 5 + 10$

Finding percents with equivalent fractions

g6_finding_percents_with_equivalent_fractions · Grade 6 – Percents · gen=percents

Name: _____ Date: _____

Find the percent.

EASY

1. What is 25% of 20?

MEDIUM

2. What is 20% of 115?

HARD

3. 361.0 is what percent of 1000?

Answer Key

1. **Easy:** 5

2. **Medium:** 23

3. **Hard:** 36.1%

Introduction to percents

g6_introduction_to_percents · Grade 6 – Percents · gen=percents

Name: _____ Date: _____

Find the percent.

EASY

1. 29 is 10% of what number?

MEDIUM

2. 84 is what percent of 112?

HARD

3. 51 is 38% of what number?

Answer Key

1. **Easy:** 290

2. **Medium:** 75%

3. **Hard:** 134.2

Relating percents, fractions, and decimals

g6_relating_percents_fractions_and_decimals · Grade 6 – Percents ·
gen=g6_relating_percents_fractions_and_decimals

Name: _____ Date: _____

Write the equivalent value.

EASY

1. Write $\frac{9}{10}$ as a decimal.

MEDIUM

2. Write $\frac{23}{50}$ as a decimal.

HARD

3. Write 0.96 as a fraction.

Answer Key

1. **Easy:** 0.9

2. **Medium:** 0.46

3. **Hard:** $\frac{24}{25}$

Solving percent problems with formulas

g6_solving_percent_problems_with_formulas · Grade 6 – Percents · gen=percents

Name: _____ Date: _____

Solve the problem.

EASY

1. 10 is what percent of 100?

MEDIUM

2. What is 5% of 180?

HARD

3. 46 is 9% of what number?

Answer Key

1. **Easy:** 10%

2. **Medium:** 9

3. **Hard:** 511.1

Kites

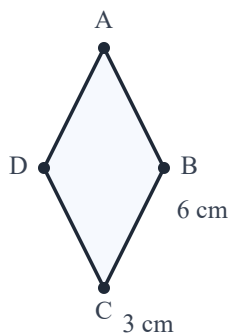
g6_kites · Grade 6 – Polygons · gen=geo_kite_area

Name: _____ Date: _____

Find the area.

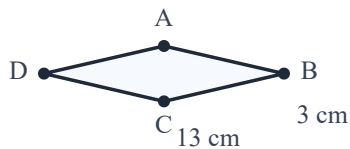
EASY

1. Find the area of the kite.



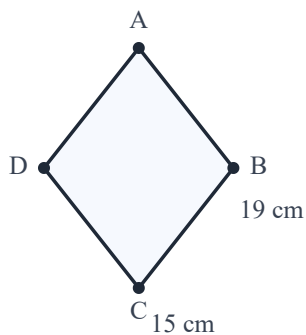
MEDIUM

2. Find the area of the kite.



HARD

3. Find the area of the kite.



Answer Key

1. **Easy:** 9 cm^2
2. **Medium:** 19.5 cm^2
3. **Hard:** 142.5 cm^2

Parallelograms

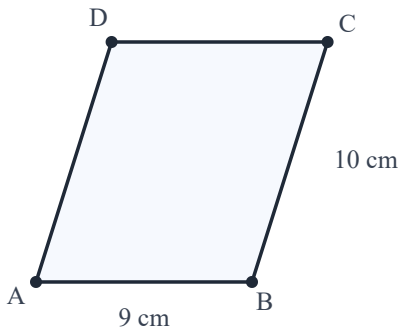
g6_parallelagrams · Grade 6 – Polygons · gen=geo_parallelagram_area

Name: _____ Date: _____

Find the area.

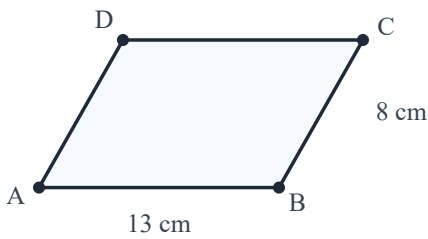
EASY

1. Find the area of the parallelogram.



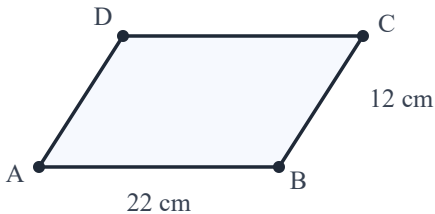
MEDIUM

2. Find the area of the parallelogram.



HARD

3. Find the area of the parallelogram.



Answer Key

1. **Easy:** 90 cm^2

2. **Medium:** 104 cm^2

3. **Hard:** 264 cm^2

Parallelograms, understanding area formula

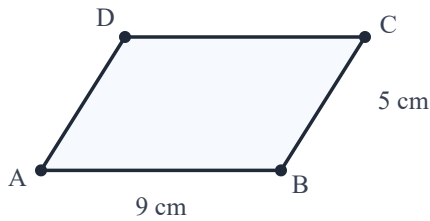
g6_parallelograms_understanding_area_formula · Grade 6 – Polygons · gen=geo_parallelogram_area

Name: _____ Date: _____

Find the area.

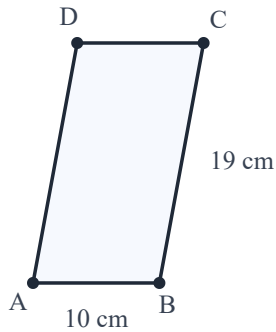
EASY

1. Find the area of the parallelogram.



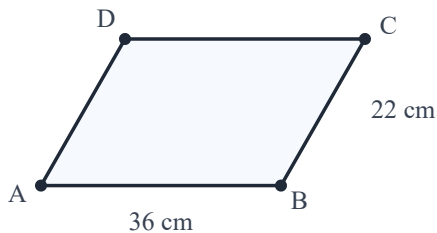
MEDIUM

2. Find the area of the parallelogram.



HARD

3. Find the area of the parallelogram.



Answer Key

1. Easy: 45 cm^2

2. Medium: 190 cm^2

3. Hard: 792 cm^2

Polygons and shaded regions

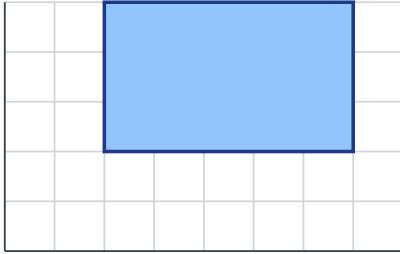
g6_polygons_and_shaded_regions · Grade 6 – Polygons · gen=g6_shaded_polygon_area

Name: _____ Date: _____

Find the area.

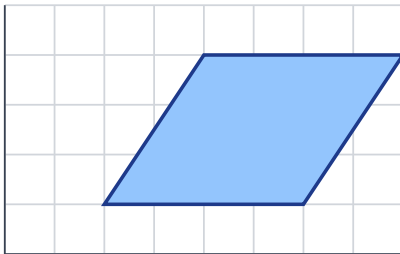
EASY

1. Find the area of the shaded region on the grid.



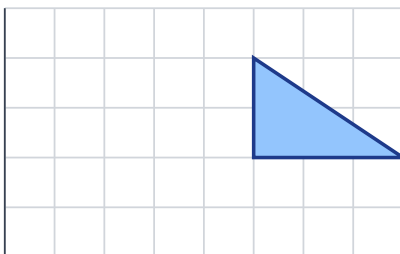
MEDIUM

2. Find the area of the shaded region on the grid.



HARD

3. Find the area of the shaded region on the grid.



Answer Key

1. **Easy:** 15 square units
2. **Medium:** 12 square units
3. **Hard:** 3 square units

Polygons on a grid or coordinate plane

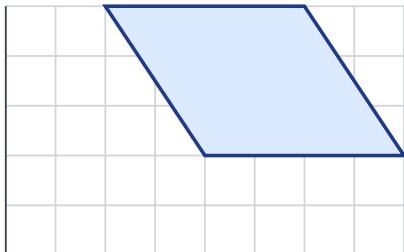
g6_polygons_on_a_grid_or_coordinate_plane · Grade 6 – Polygons · gen=g6_polygon_grid_area

Name: _____ Date: _____

Find the area.

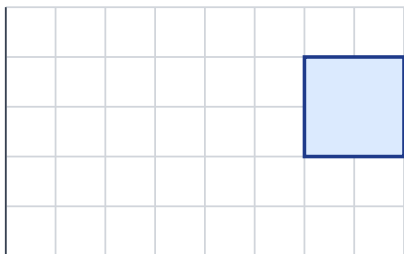
EASY

1. Find the area of the polygon on the grid.



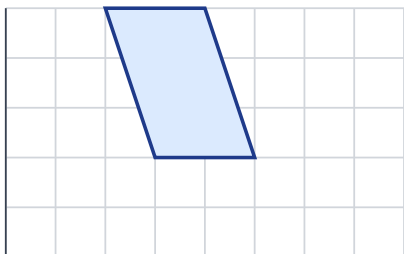
MEDIUM

2. Find the area of the polygon on the grid.



HARD

3. Find the area of the polygon on the grid.



Answer Key

1. **Easy:** 12 square units
2. **Medium:** 4 square units
3. **Hard:** 6 square units

Trapezoids

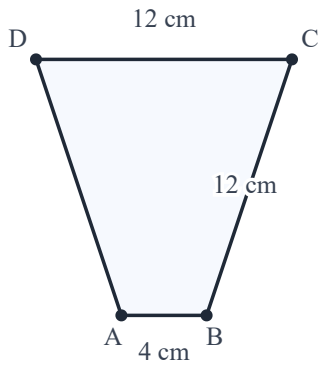
g6_trapezoids · Grade 6 – Polygons · gen=geo_trapezoid_area

Name: _____ Date: _____

Find the area.

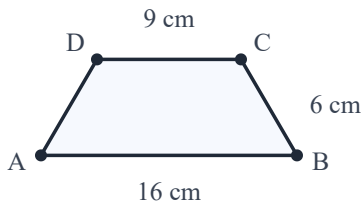
EASY

1. Find the area of the trapezoid.



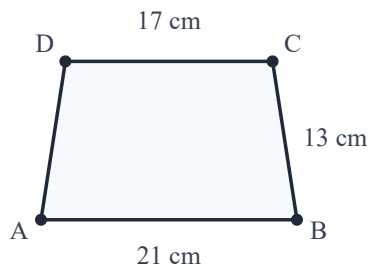
MEDIUM

2. Find the area of the trapezoid.



HARD

3. Find the area of the trapezoid.



Answer Key

1. **Easy:** 96 cm^2
2. **Medium:** 75 cm^2
3. **Hard:** 247 cm^2

Triangles

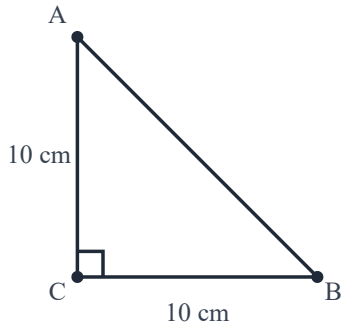
g6_triangles · Grade 6 – Polygons · gen=geo_triangle_area

Name: _____ Date: _____

Find the area.

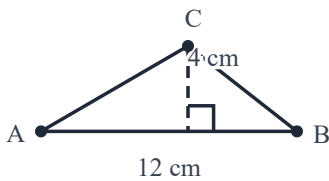
EASY

1. Find the area of $\triangle ABC$.



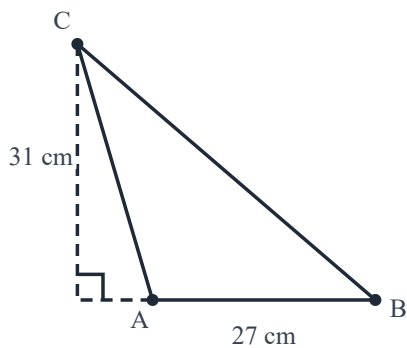
MEDIUM

2. Find the area of $\triangle ABC$.



HARD

3. Find the area of $\triangle ABC$.



Answer Key

1. **Easy:** 50 cm^2
2. **Medium:** 24 cm^2
3. **Hard:** 418.5 cm^2

Triangles, understanding area formula

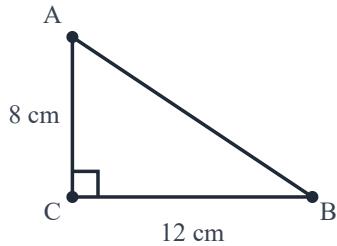
g6_triangles_understanding_area_formula · Grade 6 – Polygons · gen=geo_triangle_area

Name: _____ Date: _____

Find the area.

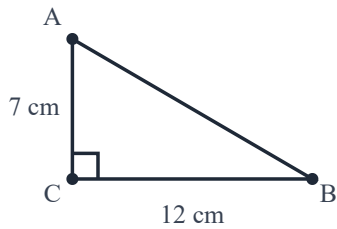
EASY

1. Find the area of $\triangle ABC$.



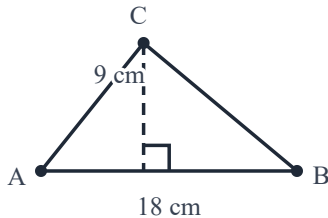
MEDIUM

2. Find the area of $\triangle ABC$.



HARD

3. Find the area of $\triangle ABC$.



Answer Key

1. Easy: 48 cm^2

2. Medium: 42 cm^2

3. Hard: 81 cm^2

Classifying and naming

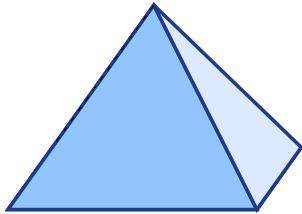
g6_classifying_and_naming · Grade 6 – Polyhedra · gen=g6_classify_polyhedron

Name: _____ Date: _____

Classify the polyhedron.

EASY

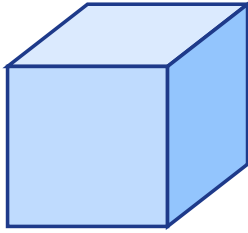
1. Classify the polyhedron shown.



- A. triangular pyramid
- B. triangular prism
- C. cube
- D. square pyramid

MEDIUM

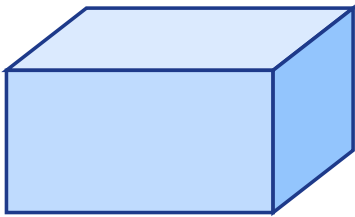
2. Classify the polyhedron shown.



- A. triangular pyramid
- B. cube
- C. square pyramid
- D. rectangular prism

HARD

3. Classify the polyhedron shown.



- A. cube
- B. rectangular prism
- C. square pyramid
- D. triangular prism

Answer Key

1. **Easy:** $D \cdot \text{square pyramid}$
2. **Medium:** $B \cdot \text{cube}$
3. **Hard:** $B \cdot \text{rectangular prism}$

Formulas for volume and surface area of a cube

g6_formulas_for_volume_and_surface_area_of_a_cube · Grade 6 – Polyhedra · gen=geo_solid_volume_surface

Name: _____ Date: _____

Find the volume or surface area.

EASY

1. Find the volume of a cylinder with radius 8 cm and height 7 cm.

MEDIUM

2. Find the surface area of a $5 \times 10 \times 9$ cm rectangular prism.

HARD

3. Find the surface area of a $36 \times 36 \times 25$ cm rectangular prism.

Answer Key

1. Easy: $64\pi \cdot 7 \text{ cm}^3$
2. Medium: 370 cm^2
3. Hard: 6192 cm^2

Volume and surface area using isometric drawings

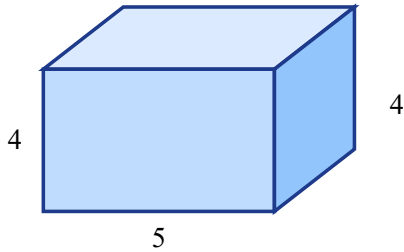
g6_volume_and_surface_area_using_isometric_drawings · Grade 6 – Polyhedra · gen=g6_isometric_measure

Name: _____ Date: _____

Find the volume or surface area.

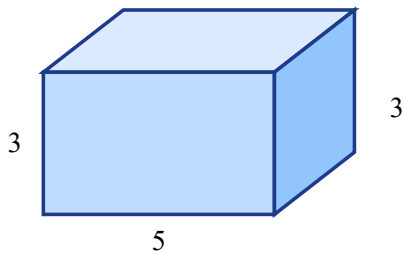
EASY

1. Use the drawing to find the surface area.



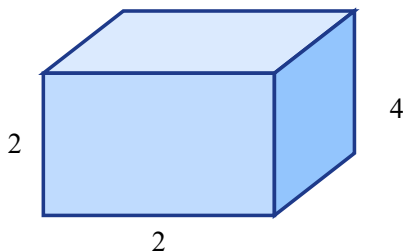
MEDIUM

2. Use the drawing to find the surface area.



HARD

3. Use the drawing to find the volume.



Answer Key

1. **Easy:** 112 square units
2. **Medium:** 78 square units
3. **Hard:** 16 cubic units

Comparing rates

g6_comparing_rates • Grade 6 – Rates • gen=g6_unit_rates

Name: _____ Date: _____

Compare the rates.

EASY

1. A student reads 18 pages in 3 minutes. Find the unit rate in pages per minute.

MEDIUM

2. \$18 for 9 pounds of fruit. Find the unit price in dollars per pound.

HARD

3. A car travels 180 miles in 9 hours. Find the unit rate in miles per hour.

Answer Key

1. Easy: 6
2. Medium: \$2
3. Hard: 20

Converting units

g6_converting_units · Grade 6 — Rates · gen=g6_unit_rates

Name: _____ Date: _____

Convert the units.

EASY

1. A car travels 15 miles in 3 hours. Find the unit rate in miles per hour.

MEDIUM

2. \$40 for 10 pounds of fruit. Find the unit price in dollars per pound.

HARD

3. \$72 for 9 pounds of fruit. Find the unit price in dollars per pound.

Answer Key

1. **Easy:** 5

2. **Medium:** \$4

3. **Hard:** \$8

Unit rates and equivalent rates

g6_unit_rates_and_equivalent_rates · Grade 6 — Rates · gen=g6_unit_rates

Name: _____ Date: _____

Find the unit rate.

EASY

1. \$9 for 3 pounds of fruit. Find the unit price in dollars per pound.

MEDIUM

2. \$15 for 3 pounds of fruit. Find the unit price in dollars per pound.

HARD

3. \$69 for 3 pounds of fruit. Find the unit price in dollars per pound.

Answer Key

1. **Easy:** \$3

2. **Medium:** \$5

3. **Hard:** \$23

Comparing ratios

g6_comparing_ratios · Grade 6 – Ratios · gen=g6_equivalent_ratios

Name: _____ Date: _____

Compare the ratios.

EASY

1. Find the missing value: $5 : 3 = 15 : x$

MEDIUM

2. $\frac{6}{11} = \frac{30}{x}$

HARD

3. Find the missing value: $6 : 19 = 12 : x$

Answer Key

1. **Easy:** 9

2. **Medium:** 55

3. **Hard:** 38

Equivalent ratios

g6_equivalent_ratios · Grade 6 – Ratios · gen=g6_equivalent_ratios

Name: _____ Date: _____

Find an equivalent ratio.

EASY

1. $\frac{1}{8} = \frac{2}{x}$

MEDIUM

2. Find the missing value: $4 : 10 = 16 : x$

HARD

3. Find the missing value: $13 : 17 = 26 : x$

Answer Key

1. **Easy:** 16

2. **Medium:** 40

3. **Hard:** 34

Introduction to ratios

g6_introduction_to_ratios · Grade 6 – Ratios · gen=g6_introduction_to_ratios

Name: _____ Date: _____

Write the ratio.

EASY

1. There are 3 blue apples and 5 purple apples. Write the ratio of blue to purple.

MEDIUM

2. There are 14 blue stickers and 15 yellow stickers. Write the ratio of blue to yellow.

HARD

3. Write the ratio 8 : 14 as a fraction in simplest form.

Answer Key

1. Easy: 3 : 5
2. Medium: 14 : 15
3. Hard: $\frac{4}{7}$

Part-part-whole ratios

g6_part_part_whole_ratios · Grade 6 – Ratios · gen=g6_introduction_to_ratios

Name: _____ Date: _____

Write the ratio.

EASY

1. There are 4 blue marbles and 1 yellow marbles. Write the ratio of blue to yellow.

MEDIUM

2. There are 14 green marbles and 6 yellow marbles. Write the ratio of green to yellow.

HARD

3. Write the ratio 2 : 11 as a fraction in simplest form.

Answer Key

1. **Easy:** 4 : 1
2. **Medium:** 14 : 6
3. **Hard:** $\frac{2}{11}$

Evaluating algebraic expressions

g6_evaluating_algebraic_expressions · Grade 6 – Variables and Algebraic Expressions ·
gen=g6_negative_number_operations

Name: _____ Date: _____

Evaluate.

EASY

1. $0 - (-7)$

MEDIUM

2. $12 - 17$

HARD

3. $40 + 38$

Answer Key

1. **Easy:** 7

2. **Medium:** -5

3. **Hard:** 78

Writing algebraic expressions

g6_writing_algebraic_expressions · Grade 6 – Variables and Algebraic Expressions ·
gen=g6_writing_algebraic_expressions

Name: _____ Date: _____

Write the expression.

EASY

1. a number decreased by 6

MEDIUM

2. 5 times the quantity of a number plus
4

HARD

3. the square of the sum of a number and 10

Answer Key

1. **Easy:** $x - 6$

2. **Medium:** $5(x + 4)$

3. **Hard:** $(x + 10)^2$